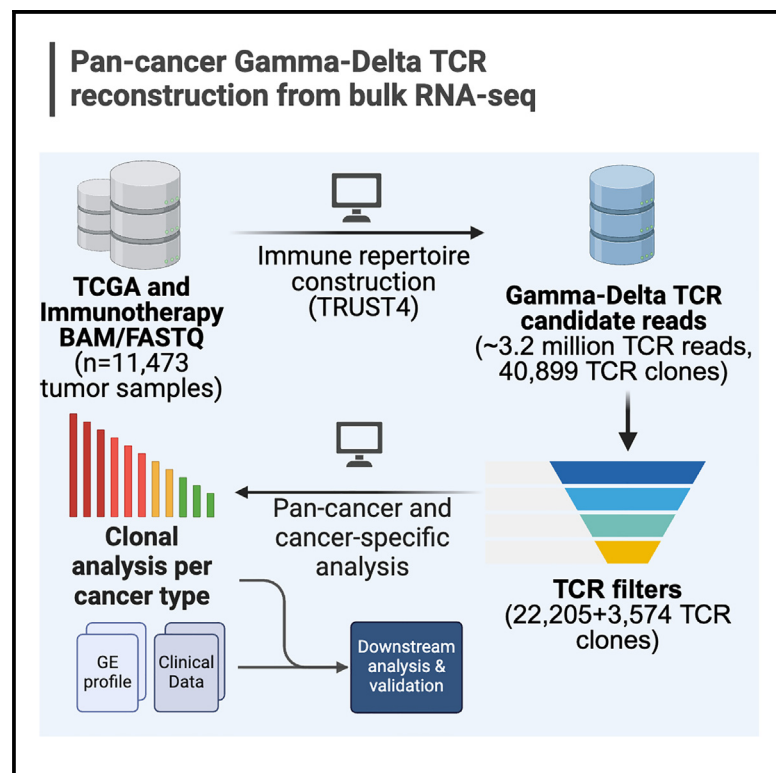


Pan-cancer $\gamma\delta$ TCR analysis uncovers clonotype diversity and prognostic potential

Graphical abstract



Authors

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In brief

Yu et al. present a large-scale analysis of gamma-delta T cell receptors spanning over 11,000 tumor samples. Their findings uncover the diversity and prognostic significance of these T cell subsets, highlighting their role in predicting cancer outcomes and offering insights for improving immunotherapy and advancing cancer research.

Highlights

- Comprehensive analysis of $\gamma\delta$ TCRs from 11,473 tumor samples
- Significant variability and overall consistency in $\gamma\delta$ gene expression and clonotype
- $\gamma\delta$ TCR expression and diversity as prognostic biomarkers across multiple cancers
- Centralized $\gamma\delta$ TCR repertoire database for future therapeutic discovery



Article

Pan-cancer $\gamma\delta$ TCR analysis uncovers clonotype diversity and prognostic potential

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SUMMARY

Gamma-delta T cells ($\gamma\delta$ T cells) play a crucial role in both innate and adaptive immunity within tumors, yet their presence and prognostic value in cancer remain underexplored. This study presents a large-scale analysis of $\gamma\delta$ T cell receptor ($\gamma\delta$ TCR) reads from 11,000 tumor samples spanning 33 cancer types, utilizing the TRUST4 algorithm. Our findings reveal extensive diversity in $\gamma\delta$ TCR clonality and gene expression, underscoring the potential of $\gamma\delta$ T cells as prognostic biomarkers in various cancers. We further demonstrate the utility of TCR gamma (TRG) and delta (TRD) gene expression from standard RNA-sequencing (RNA-seq) data. This comprehensive dataset offers a valuable resource for advancing $\gamma\delta$ T cell research, with implications for enhanced immunotherapy approaches or alternative therapeutic strategies. Additionally, our centralized database supports translational research into the therapeutic significance of $\gamma\delta$ T cells.

INTRODUCTION

Gamma-delta T cells ($\gamma\delta$ T cells) represent a specialized subset within the broader T cell population, distinguished by their T cell receptors (TCRs) that pair one gamma chain with one delta chain. Like other T lymphocyte, the diversity of $\gamma\delta$ T cells' TCRs is realized through V(D)J recombination, a process that randomly rearranges variable (V), diversity (D), and joining (J) gene segments. While constituting only a small portion of the overall T cell population, they are an important subset that uniquely contributes to both innate and adaptive immunity. Unlike their alpha-beta ($\alpha\beta$) T cell counterparts, the mechanism underlying antigen recognition by $\gamma\delta$ T cells is not yet fully elucidated.¹ The gamma-delta ($\gamma\delta$) TCRs have a significantly lower number of V genes compared to α and β chains. Yet, it is evident that they can recognize antigens in a major histocompatibility complex-unrestricted manner, enabling them to respond to a broad range of pathogens and malignant cells.² During recent years, there has been an explosive research interest in $\gamma\delta$ T cells in the realms of cancer immunology and immunotherapy, owing to their unique regulatory roles and potential antitumor functions. $\gamma\delta$ T cells, especially the V δ 1 subpopulation, have been found to infiltrate most tumors.³ On the other hand, circulating $\gamma\delta$ T cells in peripheral blood and lymphoid tissues are composed of different subpopulations, with a predominant sub-

set expressing the V δ 2 receptor.⁴ Specific subpopulations, such as V γ 9V δ 2, play a critical role in immune surveillance against infections and tumorigenesis and are present in both peripheral blood and tumor tissues. $\gamma\delta$ T cells are also known for producing various cytokines and chemokines, such as interferon gamma and tumor necrosis factor alpha. In addition, they are instrumental in a range of activities: from regulating B cells and dendritic cell maturation to recruiting macrophages and exhibiting cytolytic activity against various cancer cells.^{5–8} Recent research on both human and murine samples has revealed functional plasticity in $\gamma\delta$ T cells: the majority of $\gamma\delta$ T cells in pre-malignant or non-tumor tissues demonstrate cytotoxic effects, whereas certain tumor-infiltrating δ T cell subsets can display pro-tumorigenic effects.⁹ A more recent breakthrough discovery indicates that the antitumor efficacy of $\gamma\delta$ T cells can be potentiated or restored through targeted therapeutic approaches, specifically by targeting BTN3A1 to activate V δ 2 T cells.¹⁰ More recently, $\gamma\delta$ T cells were found associated with response to immune checkpoint inhibition in cancers with low human leukocyte antigen expression.¹¹ These discoveries open a potential avenue for treating cancers that are unresponsive to existing immunotherapy interventions.

Despite the recognized importance of $\gamma\delta$ T cells in both functional and translational research, a comprehensive understanding of their repertoire within tumor tissues remains elusive.



Pan-cancer Gamma-Delta TCR
reconstruction from bulk RNA-seq

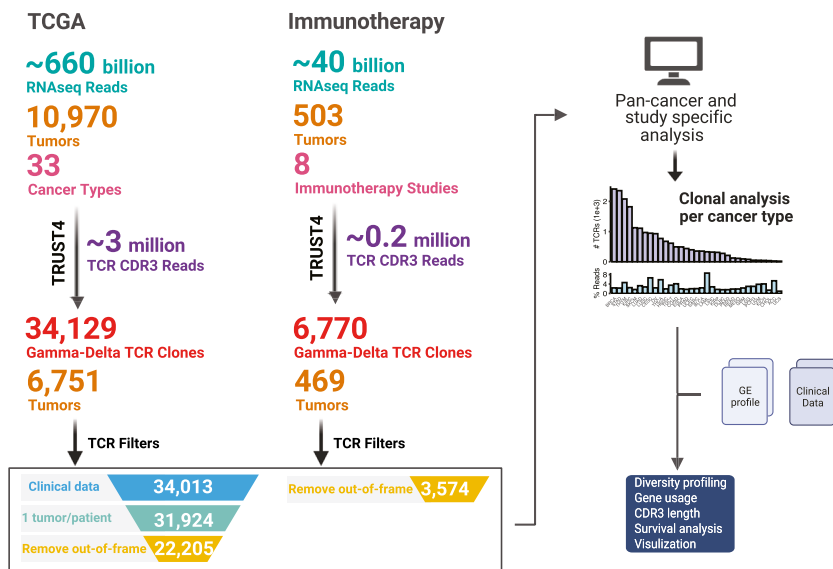


Figure 1. Comprehensive pan-cancer analysis of $\gamma\delta$ TCRs

660 million RNA-seq reads from a total of 10,970 tumor samples in the TCGA dataset were reanalyzed for $\gamma\delta$ TCRs using TRUST4. It led to the identification of 34,129 unique $\gamma\delta$ TCR clones of 6,751 tumors. $\gamma\delta$ -TCRs were further filtered to remove (1) samples without clinical information, (2) metastasis tumors for patients with multiple tumors available, (3) out-of-frame CDR3 sequences as well as CDR3 sequences carrying alpha-beta V genes. Similarly, approximately 40 million RNA-seq reads from 503 tumors downloaded from 8 immunotherapy studies were reanalyzed, generating 6,770 unique $\gamma\delta$ TCR clones, of which 3,574 clones from 459 tumors were retained for downstream analysis. Comprehensive pan-cancer and cancer/study-specific $\gamma\delta$ TCR repertoire analysis was conducted using the filtered $\gamma\delta$ TCR sequences. These refined datasets, representing a significant resource in $\gamma\delta$ TCR research, were made available to the research community, facilitating broader scientific exploration. See also Table S1.

Single-cell technologies like the 10X Chromium single-cell V(D)J enrichment solution have shed light on the diversity and clonality of T cells within tumors. However, these technologies primarily target the $\alpha\beta$ chain TCR and face limitations in sequencing depth, cost, and scalability of samples. In contrast, if repurposed for immune receptor repertoire analysis, bulk tumor sequencing offers a holistic approach for the tumor microenvironment assessment, capturing a wide spectrum of $\gamma\delta$ TCRs. Many computational tools^{12–15} have been developed to perform *de novo* assembly of the complementarity determining region 3 (CDR3) sequences using blood and tissue RNA-sequencing (RNA-seq) data. Among them, TRUST^{13,14} stands out as a sensitive and reliable method for detecting TCR CDR3 sequences from tumor RNA-seq data. Its distinct approach, which involves pairwise read comparisons, enhances the detection of the less abundant TCR clones. The results from these tools have become instrumental in elucidating the complexities of tumor-infiltrating T cell repertoires. For instance, by analyzing The Cancer Genome Atlas (TCGA) tumor RNA-seq data, TRUST was able to identify the full spectrum of T cell and B cell repertoires that infiltrate tumors across cancer types, shedding light on their probable roles in tumor immunity.¹⁶ Recently, more enhanced features have been incorporated into the upgraded algorithm, called TRUST4, which has shown better benchmarked performance.^{17,18} These computational advancements offer opportunities for addressing the significant knowledge gap concerning the less studied $\gamma\delta$ T cell repertoire—via revisiting pan-cancer datasets.

In this report, we reanalyzed raw RNA-seq data from nearly 11,000 tumor samples utilizing the latest version of TRUST4. This allowed us to offer a highly detailed and comprehensive examination of the $\gamma\delta$ TCR landscape spanning 33 different cancer types, providing one of the most extensive collections of $\gamma\delta$ TCR

reads in human cancer. A series of downstream analyses were performed to further investigate the pan-cancer repertoire landscape, the association between $\gamma\delta$ TCR clones and VDJ gene expression, as well as their implications in patient prognosis. Collectively, our results and findings serve as a foundational resource for $\gamma\delta$ T cell research in oncology.

RESULTS

Reconstructing $\gamma\delta$ TCRs from 11,473 tumor RNA-seq samples

We utilized the computational method TRUST4 to process RNA-seq data, encompassing 10,970 BAM files, from 10,131 patients within TCGA database (Figure 1; Table S1). The vast majority of these samples consist of paired-end RNA-seq data, except for 637 samples which are single-end. This updated immune repertoire reconstruction tool allowed for the efficient and sensitive *de novo* assembly of CDR3 sequences derived from TCR transcripts. Through this computationally intensive effort, our objective was to curate the most comprehensive pan-cancer catalog of $\gamma\delta$ TCRs across all cancer types represented in TCGA. All analyses presented in this study were based on TCGA TCR assembly data that were fully completed in January 2024. From the entire TCGA RNA-seq dataset, we extracted approximately 3 million reads covering the TCR CDR3 region. These reads were annotated with information such as amino acid sequences and associated V and J genes. To reduce false positive calls, we applied a series of stringent filtering criteria—by removing biologically implausible and ambiguous sequences, e.g., unproductive CDR3s and those concurrently annotated as $\alpha\beta$ TCRs. This process led to the identification of 34,129 unique $\gamma\delta$ TCR clones. Among the analyzed tumors, 6,751 tumors yielded at least one copy of a $\gamma\delta$ TCR clone. In total, when considering only one tumor sample per patient, we identified 31,924 unique clones. After further

filtering to exclude out-of-frame CDR3s, this number was refined to 22,205 unique clones (Figure 1).

In addition to the TCGA pan-cancer dataset, we also downloaded and processed raw RNA-seq data of a total of 503 tumors from 8 immunotherapy studies (see STAR Methods). Similar to TCGA dataset, we performed TRUST4 analysis and extracted approximately 0.2 million reads covering the TCR CDR3 region. This process led to the identification of 6,770 unique $\gamma\delta$ TCR clones.

A pan-cancer catalog of $\gamma\delta$ TCRs reveals significant variability across cancers

We explored the gamma chain expression and clonotype count information across 33 distinct cancer types (Figure 2A). TCR gamma gene expression for each sample is represented by an enrichment score, calculated based on single-sample gene set enrichment analysis on all TCR gamma genes (e.g., TRGV9). We observed significant variability in enrichment scores, both between and within cancer types (Figure 2A top panel). As expected, tumors traditionally classified as immune-cold cancers,¹⁹ such as UVM and GBM, demonstrate the lowest enrichment scores, while lymphatic system-related cancers like LAML, THYM, and DLBC have the highest. Beyond THYM, the top five solid-tumor cancer types carrying the highest enrichment are PRAD, KIRC, TGCT, LUAD, and KICH. We next examined the number of patients that have at least one copy of a TCR gamma clone and the number of patients with no reconstructed functional TCR gamma clone for each cancer type (Figure 2A second panel). In cancers that rank highest in gamma gene enrichment (spanning from LAML to STAD), cancers generally have a high proportion of patients with detectable TCR gamma clones, with PRAD being a notable exception. Among the intermediate-ranked cancers, such as BRCA, HNSC, and THCA, approximately 50% of patients have detectable gamma clones. The lower ranked cancers follow expected trends of fewer clones, but with exceptions like OV. We investigated the number of TCR gamma clones along with the clone size distribution across cancer types (Figure 2A third and bottom panels). Overall, a dominant presence of large clone segment is not observed in most cancers. Cancers including LAML, KIRC, LUAD, SKCM, CESC, LUSC, STAD, SARC, HNSC, ESCA, OV, and BLCA do display a relatively higher proportion of intermediate to larger clones compared to other cancer types. An analysis of delta gene expression and clones reveals a similar pattern (Figure S1).

Compared to their gamma gene counterparts, delta gene expression and clone frequency are notably sparser across various cancer types. It is important to note that these clone sizes should be interpreted as downsampled relative frequencies, considering the small proportion of $\gamma\delta$ TCRs and the fact that we are repurposing bulk RNA-seq for immune repertoire analysis. The actual clone sizes are anticipated to be much larger when profiled using a targeted gamma chain TCR assay.

We further checked the distribution of out-of-frame (OOF) TCR reads (Figures S2A and S2B). Although OOF TCRs do not contribute to functional immune response, the ratio of in-frame/OOF TCRs could serve as a biomarker, and these sequences might provide insights into the diversity of progenitor T cells. However, we did not observe any cancer-specific pat-

terns in terms of in-frame/OOF ratios (Figure S2A). The range of these ratios in gamma chain was rather uniform across different cancer types, varying between 27% and 41%. The overall distribution of OOF clone abundance mirrored that of in-frame reads, with the highest yields observed in STAD and BRCA. Most of these clones are singleton counts, suggesting that they are indeed non-functional. A similar pattern was observed in TCR delta chains, albeit with a significantly sparser distribution.

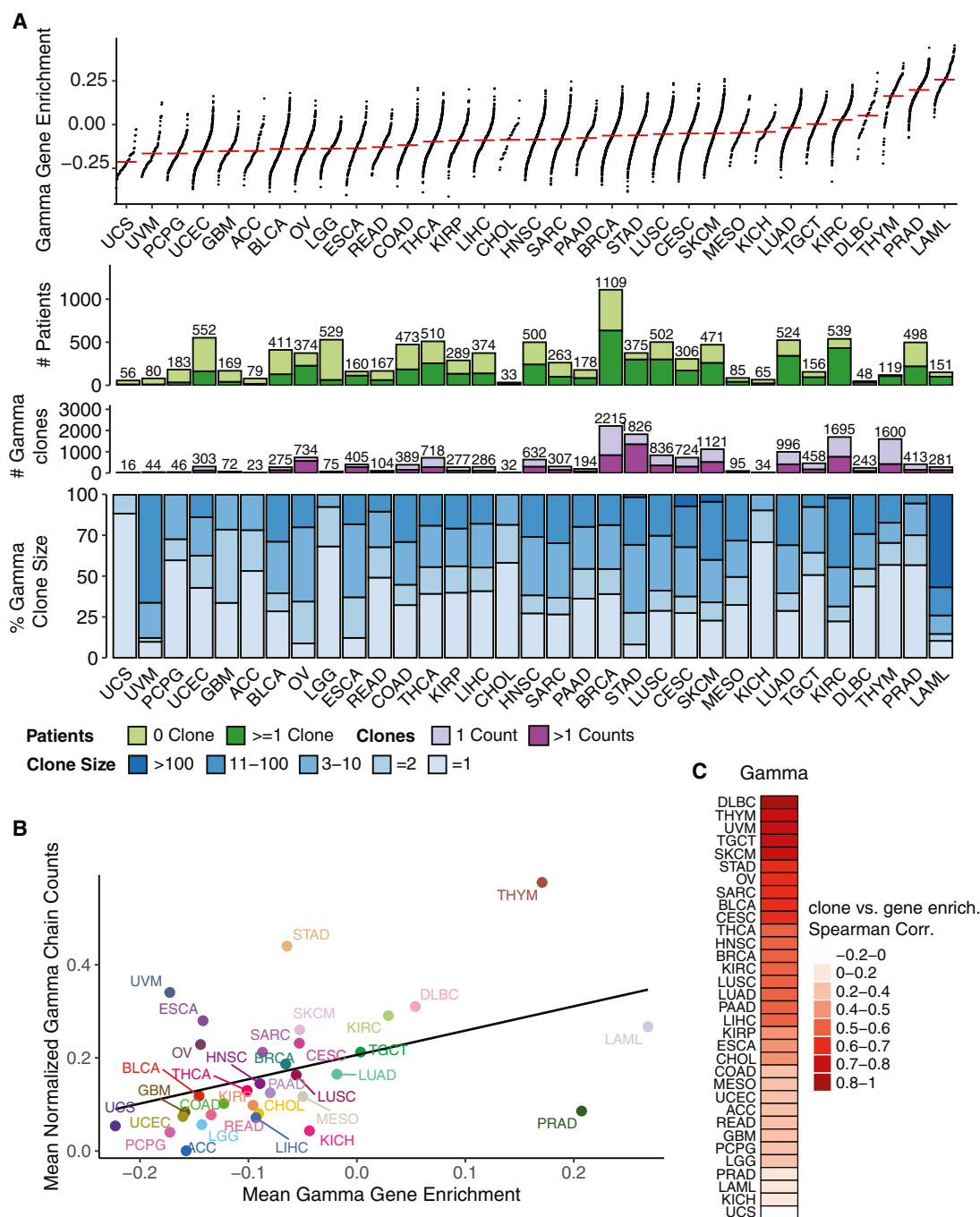
In our data, the CDR3 length distribution of $\gamma\delta$ TCRs is in line with the patterns in beta chains,¹³ where the lengths of these CDR3 regions predominantly range from 10 to 15 amino acids (Figure S2C). Overall, the delta chain tends to have a slightly longer region compared to the gamma chain, which could be due to the inclusion of D genes. When we rank cancer types by the length of the CDR3, there is a noticeable inconsistency between the rankings for gamma and delta chains.

Overall consistency between $\gamma\delta$ gene expression and clonotype

Next, we conducted a comparison between the mean gamma gene enrichment score with the mean normalized gamma chain counts across the 33 cancer types and observed an intermediate positive correlation pattern (Figure 2B). Several cancers, such as STAD and THYM, are observed to slightly deviate above the trend line, suggesting higher TCR gamma reads than what might be expected from the gene expression-based enrichment score. Cancers such as BRCA, TGCT, HNSC, and BLCA align closely to the trend line, indicating a more direct relationship between their gene expression and TCR gamma readouts. Conversely, PRAD stands out as a notable outlier below the trend line, emphasizing that the gene expression might not always mirror the total abundance or diversity of productive CDR3 sequences. This pattern highlights the complexity of the relationship between gene transcription levels and the resulting functional immune repertoire. For a more in-depth sample-wise analysis, we further examined the relationship between normalized gamma CDR3 counts and both gamma and delta gene expression enrichment within each cancer type, accounting for tumor purity (Figure S1B). About two-thirds of the cancers exhibited intermediate to strong correlations, confirming a generally predictable relationship between gene expression and TCR gamma reads for the majority of cancer types examined. In addition to the normalization chain counts, we calculated the clonality metrics for all cancer types (Figures S3A and S3B). Our analysis revealed a broad spectrum of clonality, with values ranging from 0 to 1 (larger values mean more clonal expansion), across patients in all cancer types. Furthermore, it was noted that the most immunogenic cancers, specifically SKCM, BLCA, and HNSC, generally display higher median clonality.

$\gamma\delta$ TCR V-J gene pair distribution across cancer types

The landscape of $\gamma\delta$ TCR V-J gene pair distribution offers another layer of insight into clonality and the dynamics of $\gamma\delta$ immune repertoire across different cancer types (Figures S3C and S3D). In the gamma V-J pair distribution, a notable pattern emerges with the gene pair TRGV10_TRGJ1 being consistently dominant across all cancer types (Figure S3C). Notably, other prevalent



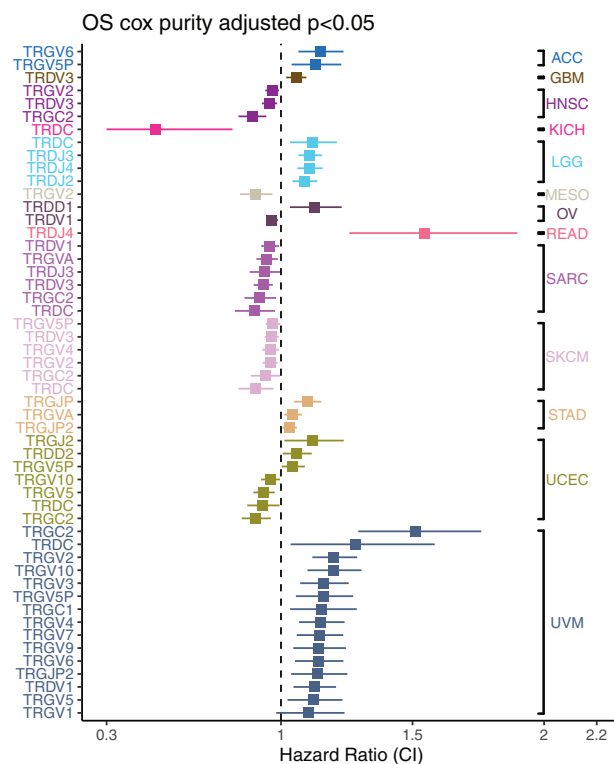


Figure 3. Prognostic implications of $\gamma\delta$ TCRs in TCGA

The forest plot displays the results of the Cox regression survival analysis on TCR γ and δ gene expression adjusted by covariates across various cancer types. It highlights genes with significant or marginally significant impacts on patient survival, indicated by FDR-adjusted Cox regression p values below 0.05. Each horizontal line in the forest plot represents a γ or δ gene in a cancer type, with the square and length of the line indicating the estimated hazard ratio (HR) and its 95% confidence interval, respectively. Horizontal lines are colored by cancer types. Within each cancer type, genes are ranked by HR from highest to lowest. A dashed vertical line at HR = 1 represents no effect.

gene pairs also involve TRGJ1, including TRGV9_TRGJ1, TRGV2_TRGJ1, TRGV4_TRGJ1, TRGV3_TRGJ1, and TRGV8_TRGJ1. Remarkably, in all cancer types examined, the top 5 most prevalent gene pairs account for over 75% of all $\gamma\delta$ TCR reads, suggesting a shared positive selection mechanism during T cell development. In the delta V-J pair distribution, as illustrated in Figure S5B, TRDV1_TRDJ1 stands out as the dominant gene pair. Other notable pairs include TRDV2_TRDJ1, TRDV1_TRDJ2, and TRDV3_TRDJ1 (Figure S3D). However, the diversity observed in delta V-J pairs is much lower compared to their gamma counterparts, likely due to the reduced number of reads obtained from the delta chain, which may result from lower expression of the delta chain or the frequent deletion of the TCR delta (TRD) locus following TCR gene rearrangement.²⁰

Prognostic implications of $\gamma\delta$ TCRs in the pan-cancer dataset

To further explore the prognostic potential of tumor-infiltrating $\gamma\delta$ T cells, we performed a systematic survival-analysis screening focused on the gene expression levels of TCR gamma and delta genes across various cancer types (Figure 3). We listed all genes that showed either a significant or marginal impact on patient overall survival, as indicated by their Cox regression p values (false discovery rate [FDR] adjusted) being

less than 0.05. In performing the Cox regression analysis, we adjusted for covariates including gender, age, stage, and tumor purity. Please note that in our study, the Cox regression and corresponding p values were utilized primarily to prioritize the $\gamma\delta$ TCR genes for further discussion, rather than to build predictive models or to establish causal relationships. Our analysis identified 24 $\gamma\delta$ TCR genes as potential prognostic biomarkers in 13 cancer types. Among them, four genes (*TRGC2*, *TRGV2*, *TRDC*, and *TRDV3*) appeared in more than 3 cancer types, with *TRDC*, often serving as a proxy for the overall presence of $\gamma\delta$ T cells,²¹ being notably present in 5 different cancers. Another interesting gene, *TRDV3*, a biomarker for the $\gamma\delta 3$ subset, has recently attracted attention due to its emerging significance in immune cancer immunology.¹¹ As expected, the expression of these in cancers like SKCM and HNSC showed an association with favorable survival, contrasting with varied directions in less immunogenic cancer types.

Acknowledging the complexity inherent in transcriptome-based prognostic signatures, we specifically examined HNSC and COAD, two cancers known for their distinct tumor subtypes (Figure 4). HNSC is categorized into HPV-positive and negative subtypes, each with a unique immunological landscape and clinical outcomes. Similarly, COAD exhibits distinct molecular and clinical variability between microsatellite instability (MSI)

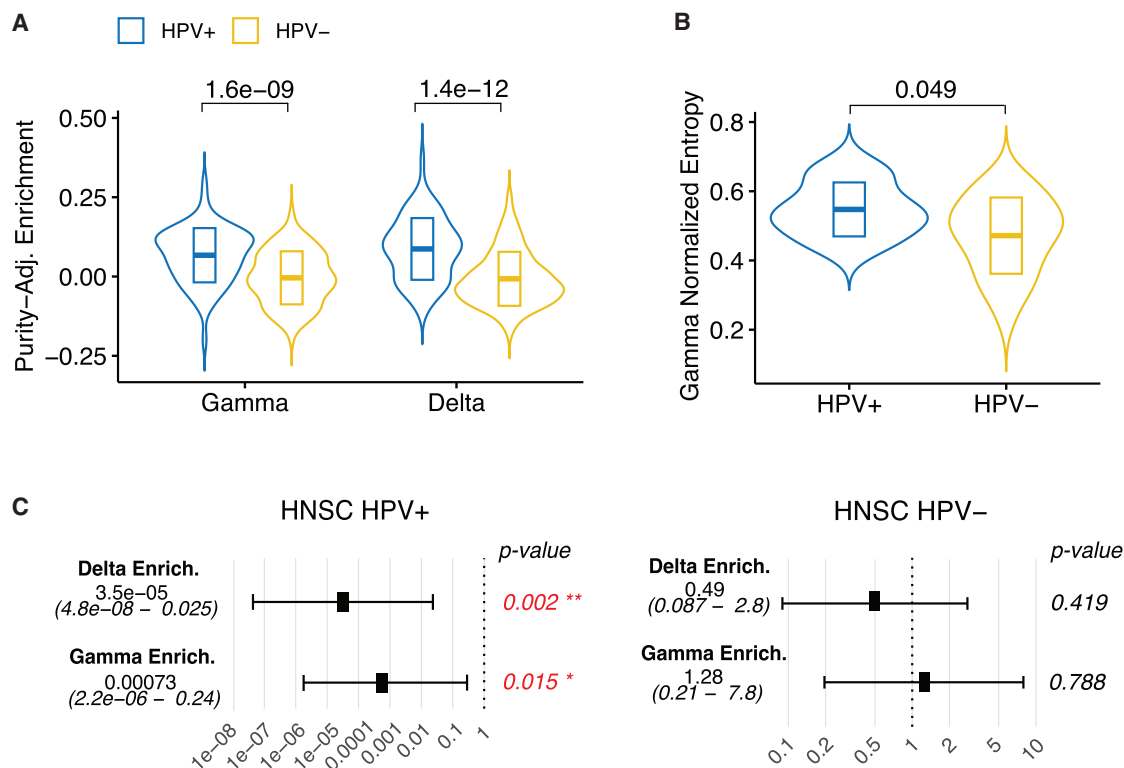


Figure 4. $\gamma\delta$ TCR-based prognostic signatures in HPV-positive vs. HPV-negative HNSC tumors in TCGA

(A) Elevated TCR gamma and delta gene expression in patients with HPV-positive HNSC. Y axis represents the purity-adjusted gamma (left) and delta (right) gene enrichment scores.

(B) Higher normalized entropy of TCR gamma signature in the HPV-positive HNSC. Forest plots show the estimated hazard ratio (square) and its 95% confidence interval (horizontal line) of gamma and delta gene enrichment by Cox regression adjusted for covariates in HPV+ and HPV- HNSC, respectively. A dashed vertical line at HR = 1 represents no effect.

(C) Association of gamma and delta TCR signatures with improved survival in HPV-positive HNSC, absent in HPV-negative. *p* value was calculated by two-sided Wilcoxon rank-sum test. Violin plot shows the kernel probability density of the data with box in middle representing the median (central line), the 25% and 75% interquartile (IQR) (lower and upper hinges).

See also Figures S4A–S4D.

and microsatellite stable subtypes. Both TCR gamma and delta signatures are notably higher in patients with HPV-positive HNSC, suggesting a more active or diverse $\gamma\delta$ T cell presence in these patients (Figure 4A). Similarly, the normalized entropy of the TCR gamma signature, a measure indicative of gamma TCR diversity, was also found to be higher in the HPV-positive patient group (Figure 4B). Interestingly, both gamma and delta TCR signatures were significantly associated with patient survival, exhibiting a protective effect in the HPV-positive HNSC subgroup (Figure 4C). However, this association was not observed in the HPV-negative patient group, suggesting that the presence of $\gamma\delta$ T cells might contribute differently to the disease progress in these subtypes. We also conducted a detailed comparative analysis of HPV-positive and HPV-negative samples, focusing on individual genes, gamma clonality, and genes associated with survival in the stratified groups (Figures S4A–S4D). The analysis indicates that patients with HPV-positive typically exhibit a more diverse range of TCR gamma reads. We also observed notable disparities in the prognostic implications of TRG and TRD genes, with these genes appearing more prognostic in the HPV-positive group.

In the context of COAD, we noted significantly higher values in both gamma and delta TCR enrichment scores in the MSI-high subtype of COAD, alongside an elevation in the normalized entropy of the gamma TCR, indicating a more diverse T cell repertoire in the MSI subgroup (Figures 5A and 5B). There were differences in the TRG and TRD gene usage and gamma clonality between the two COAD subtypes (Figures S4E–S4G). Intriguingly, we observed that the gamma enrichment score is significantly associated with worse survival outcomes in the MSI-low group of COAD, contrasting with its null association in the MSI-high group (Figure 5C). This disparity reflects the complex and multifaceted role of $\gamma\delta$ T cells in influencing cancer prognosis.

Prognostic implications of $\gamma\delta$ TCRs in immunotherapy-treated tumors

We extended our analysis of $\gamma\delta$ TCRs beyond the TCGA pan-cancer dataset by incorporating RNA-seq data from 503 tumors across 8 immunotherapy studies. This additional data allowed us to investigate the prognostic significance of $\gamma\delta$ TCR signatures in the context of immunotherapy. In the study by Gide (Figure 6A),

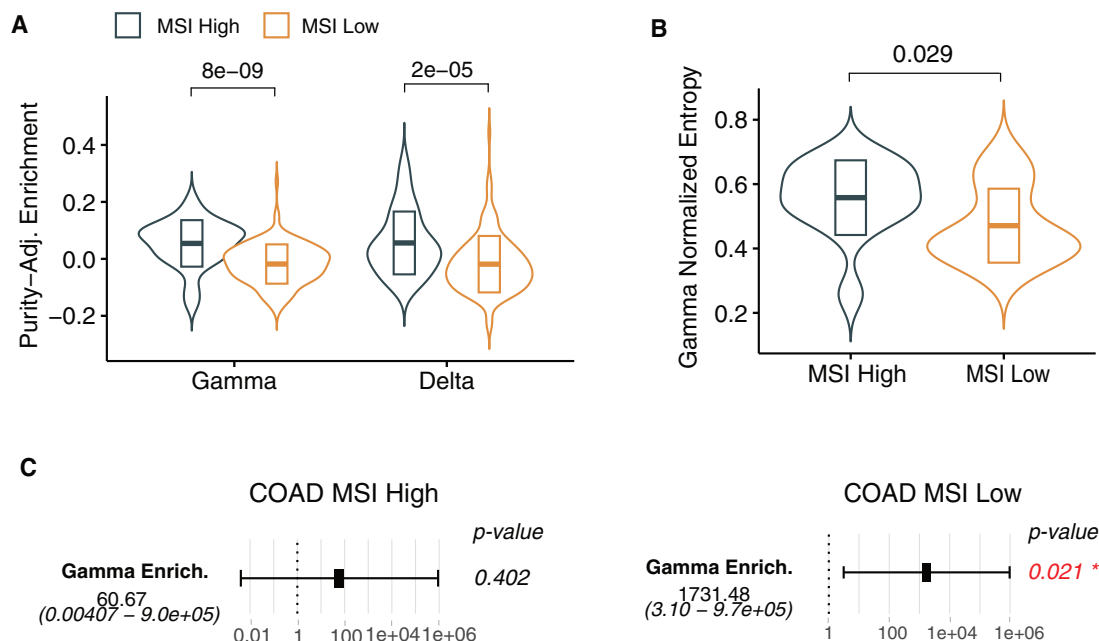


Figure 5. $\gamma\delta$ TCR-based prognostic signatures in TCGA COAD tumors

(A) Significantly higher gamma and delta TCR enrichment scores in the MSI-high group.

(B) Higher normalized entropy of the gamma TCR in the MSI-high group.

(C) The gamma enrichment score associated with worse survival outcomes in the MSI-low group. Forest plots show the estimated hazard ratio (square) and its 95% confidence interval (horizontal line) of gamma gene enrichment by Cox regression adjusted for covariates in MSI-high and MSI-low COAD tumors, respectively. A dashed vertical line at HR = 1 represents no effect. For (A) and (B), p value was calculated by two-sided Wilcoxon rank-sum test. Violin plot shows the kernel probability density of the data with box in middle representing the median (central line), the 25% and 75% interquartile (IQR) (lower and upper hinges). See also Figures S4E–S4G.

we observed that the enrichment of gamma and delta genes was higher in responders (complete response/partial response) compared to non-responders (stable disease/progressive disease) in both pre-treatment and on-treatment cohorts, and in the overall patient cohort (Figure S5A). Additionally, higher gamma gene enrichment scores were associated with improved survival outcomes in both pre-treatment (middle panel in Figure 6A) and on-treatment samples (Figure S5B). Specifically, three individual genes (TRGV2, TRGV8, and TRGV10) demonstrated a protective effect, as shown by the Cox regression analysis of progression-free survival, adjusted for age and gender. In the study by Riaz (Figure 6B), we observed a similar pattern across the entire cohort (left panel) and in the ipilimumab-progressive (Ipi-Prog) patients, but not in the ipilimumab treatment-naïve (Figure S5C). Within the Ipi-Prog subgroup (right panel), gamma and delta gene enrichment scores were higher in responders when stratified by pre-treatment and on-treatment tumors. The Kaplan-Meier overall survival plot (Figure 6C) also indicated that patients with higher gamma enrichment scores in pre-treatment samples tended to have better survival outcomes; a similar pattern was observed in on-treatment samples (Figure S5D).

We further analyzed the diversity metrics of gamma chains in immunotherapy studies with patient response information (Figure S5E). Our findings indicate that responders, or patients who benefited from treatment, exhibited a more diverse gamma

TCR repertoire compared to non-responders across multiple studies. In the study by Riaz, we observed that the gamma diversity score, measured by normalized entropy, was associated with improved patient survival in both pre-treatment and on-treatment samples (Figure 6D). A similar pattern was observed in the pre-treatment cohort of the study by Gide (Figure S6F).

Although based on limited data, these findings confirm the prognostic value of gamma and delta gene expression enrichment and diversity metrics in immunotherapy. Incorporating these $\gamma\delta$ TCR signatures could enhance the precision of patient stratification and treatment personalization when used alongside other established biomarkers.

Development of an interactive $\gamma\delta$ TCR webtool

To facilitate the characterization of $\gamma\delta$ TCR repertoires, we have developed an open-access resource and web server <http://gdt.moffitt.org>. This resource curates all $\gamma\delta$ repertoires reconstructed with TRUST4, along with gene expression data of $\gamma\delta$ TCR genes and clinical information from both TCGA and immunotherapy studies. The web server includes three major modules. (1) *TCR search*. This module allows users to query $\gamma\delta$ TCRs identified in both TCGA and immunotherapy studies through CDR3 amino acid sequences (with mismatches allowed) and V/D/J genes. All TCRs matching the querying criteria will be displayed along with their abundance and associated clinical information. (2) *Gene expression analysis*. In this module, users

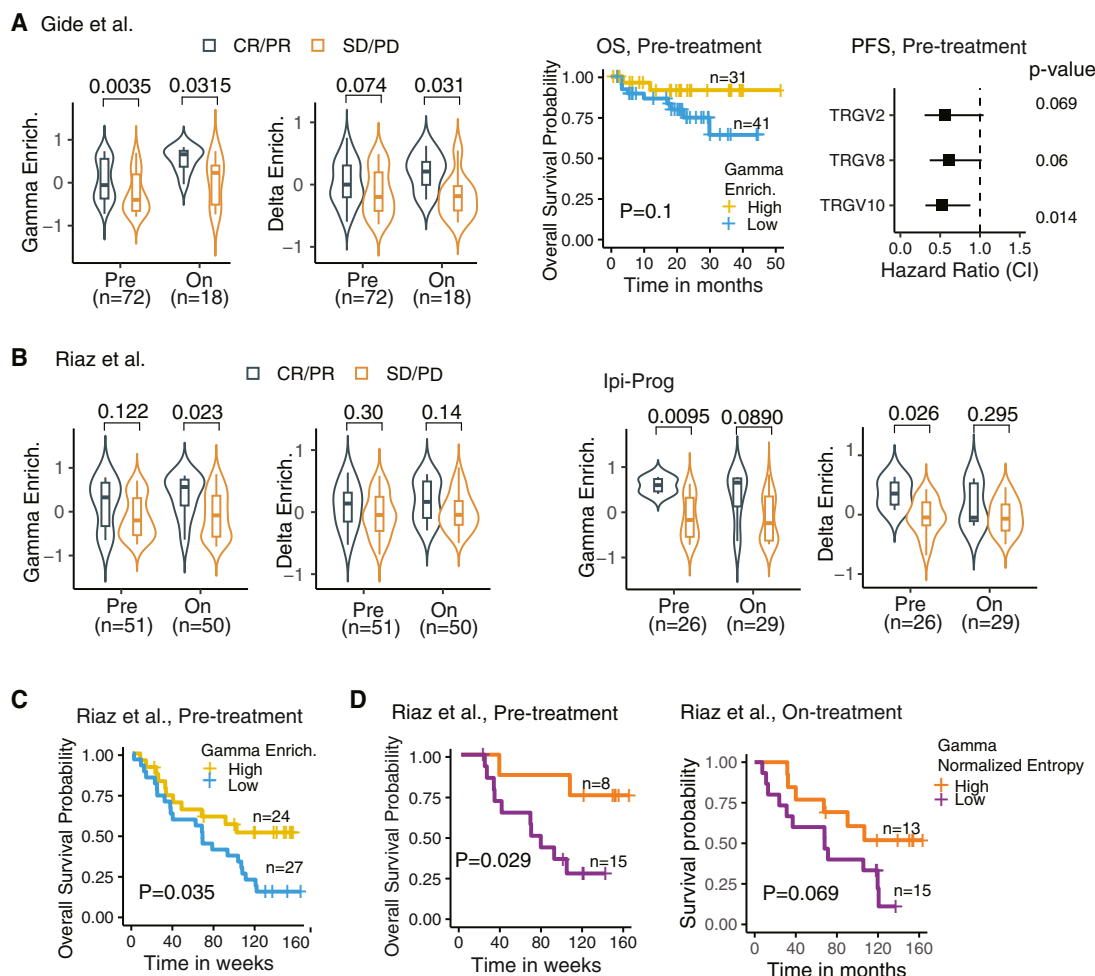


Figure 6. $\gamma\delta$ TCR-based prognostic signatures in immunotherapy studies

(A) Prognostics $\gamma\delta$ TCR signatures in the study by Gide. Violin plots on left compare gamma and delta gene enrichment between patients with complete response (CR) or partial response (PR) vs. patients with stable disease (SD) and progressive disease (PD), in pre-treatment and on-treatment samples, respectively. Number of samples in each category was labeled below x axis. Right: the (1) Kaplan-Meier overall survival plot by patients' pre-treatment samples stratified based on optimal cutoff of gamma gene enrichment, with yellow representing high and blue representing low. (2) Forest plot of PFS (progression-free survival) HR (square) and its 95% confidence interval (horizontal line) of gamma gene enrichment by Cox regression adjusted for age and gender. A dashed vertical line at HR = 1 represents no effect. *p* values are shown on right.

(B) Prognostics $\gamma\delta$ TCR signatures in the study by Riaz. Left: violin plot with box shows elevated gamma and delta gene enrichment in patients with CP/PD vs. patients with SD/PD, in both pre- and on-treatment tumors. Right: violin plot with box shows elevated gamma and delta gene enrichment in CP/PD vs. SD/PD, in both pre- and on-treatment tumors within ipilimumab treatment-progressive (Ipi Prog) patients.

(C) In the study by Riaz, Kaplan-Meier overall survival plot by patients' pre-treatment samples shows that higher gamma gene enrichment is associated with better survival. Patients are stratified based on optimal cutoff of gamma gene enrichment.

(D) Kaplan-Meier overall survival plot by patients' pre-treatment samples stratified based on optimal cutoff of gamma true diversity in the study by Gide (left) and gamma normalized entropy in the study by Riaz (right), with orange representing high and purple representing low. For violin plots in A–C, *p* value was calculated by two-sided Wilcoxon rank-sum test. Violin plot shows the kernel probability density of the data with box in middle representing the median (central line), the 25% and 75% interquartile (IQR) (lower and upper hinges), and the ± 1.5 IQR (Tukey whiskers).

See also Figure S5.

can associate the expression of individual $\gamma\delta$ TCR genes with various clinical outcomes through hierarchically clustered heat-maps, in both TCGA and immunotherapy studies. (3) *Prognostic analysis*. Survival analysis (3A): this sub-module provides a comprehensive evaluation of the prognostic significance of features including individual $\gamma\delta$ TCR gene expression, gamma/delta gene overall enrichment, and gamma/delta clone diversity.

When a feature is queried in a specific cancer type or immunotherapy study, the module renders a Kaplan-Meier plot with either optimal or median cutoff by the choice of users and a forest plot showing the hazard ratio and its confidence interval from a Cox regression adjusted for all covariates. Users can select the desired maximum length of survival time for both Kaplan-Meier and Cox regression analysis. Association with clinical outcomes

(3B): this sub-module enables users to explore the $\gamma\delta$ gene expression as well as gamma/delta clone diversity with different clinical outcomes, such as responders vs. non-responders, pre- vs. post-treatment, live vs. dead, and various treatment combinations. By providing these functions, this web tool aims to support ongoing research and therapeutic discovery by providing easy access to extensive $\gamma\delta$ TCR data and associated clinical insights.

DISCUSSION

Despite being a minority group in the T cell community, $\gamma\delta$ T cells have emerged as a critical area of interest in cancer immunology and translational research. Their dual capability to mirror both innate and adaptive immune responses positions $\gamma\delta$ T cells as promising candidates for clinical biomarkers, aiding in the tracking of cancer progression and response to various treatments, including immunotherapy.^{10,21–24} Additionally, their complementary immunosurveillance role alongside $\alpha\beta$ T cells underscores their potential as therapeutic targets in combination therapies.²⁵ Most of previous studies indicate a protective value for V δ 1+ $\gamma\delta$ T cells, but the impact of specific gamma chains that they express and the roles of $\gamma\delta$ tumor-infiltrating lymphocytes differing from V δ 1 or V δ 2 on their function remain unclear.

Our comprehensive analysis of the $\gamma\delta$ TCR landscape across 33 cancer types, utilizing the TRUST4 algorithm, provides key insights into the role and potential utility of $\gamma\delta$ T cell signature in different cancer tissues. This study represents a highly detailed and comprehensive collection of $\gamma\delta$ TCR reads in human cancer, providing a critical resource for the ongoing intensive research into this vital immune component. The yielded database offers applications across many research scenarios. For example, users could use it to query the candidate TCR sequences and determine the degree of tumor and tissue specificity. Additionally, they can also compare these sequences against the cancer-specific dataset to identify the common clones or conduct distance-based clustering analysis.²⁶ Although the variation in CDR3 lengths across cancer types might not provide immediate biological insights, the large collection of the $\gamma\delta$ TCR repertoire serves as a valuable reference database. This enables users to detect changes in length distribution (e.g., TCR spectratyping) in individual patients' TCR repertoires and monitor $\gamma\delta$ -specific clonal expansion or selection. Our analysis also highlights the often-overlooked utility of TRG and TRD gene expression values, readily obtainable through standard RNA-seq data analysis. These values offer alternative methods for not only assessing the overall activity of $\gamma\delta$ T cells but also for uncovering specific genes that could serve as potential gene-level biomarkers. For instance, in our prior research, we identified *TRGV9* transcripts as key biomarkers in prostate cancer.²⁷ It is important to acknowledge that expression values of TRG and TRD genes are available in the most current reference annotations for the human genomes, such as GENCODE, but are absent in older versions of the RefSeq gene annotation. Our findings that $\gamma\delta$ TCR clones are prevalent across all cancer types, particularly the dominance of specific V-J gene pairs, hint at a possible link to common antigens or tumor immune response mechanisms. The fact that most immunogenic cancers tend to have higher median TCR $\gamma\delta$ gene

enrichment scores and clonality further supports the idea of a locally converged immune response in these cancers, possibly against specific tumor antigens.

Recent findings indicate that although, in patients with cancer, $\gamma\delta$ T cell presence correlates with better outcomes, intra-tumor $\gamma\delta$ T cells can exhibit a temporal pattern in which their subsets play a different role in supporting antitumor immunity.⁹ Consequently, it is important to emphasize that any static analysis of $\gamma\delta$ TCR clonality or gene expression for prognostic purposes (in testing for the association with patient survival outcomes) should be approached with caution. With this caveat noted, our comprehensive survival analysis has identified multiple TRG and TRD genes as potential prognostic biomarkers across multiple cancer types. Our results in HNSC and COAD also indicate that the role of $\gamma\delta$ T cells in disease progression and patient prognosis varies based on the molecular characteristics of the tumor. In HPV-positive HNSC, the higher diversity and clonality of $\gamma\delta$ TCRs, and their association with improved survival, contrast with the lack of such associations in HPV-negative HNSC. Likewise, in COAD, the distinct patterns observed between MSI-high and MSI-low subtypes in terms of $\gamma\delta$ TCR enrichment and their association with survival outcomes further emphasize the complex, context-specific contribution of $\gamma\delta$ T cells in cancer progression.

Our extended analysis of $\gamma\delta$ TCRs in immunotherapy datasets, including RNA-seq data from 503 tumors across 8 studies, underscores the significant role of $\gamma\delta$ T cells in predicting patient response to immunotherapy. In multiple studies, we observed that patients with immunotherapy benefits tend to exhibit higher gamma and delta gene enrichment scores, as well as higher diversity in gamma chain TCRs. These results suggest that incorporating $\gamma\delta$ TCR signatures could further enhance the precision of treatment personalization in immunotherapy. Although our findings highlight the prognostic value of intra-tumor $\gamma\delta$ T cells and specific TRG and TRD genes, we must acknowledge that the true therapeutic implications of $\gamma\delta$ T cells are multifaceted. Different $\gamma\delta$ T cell subsets are activated by distinct stimuli, such as *BTNL3* and *BTNL8* for V γ 4V δ 1 cells,^{28,29} which suggests that tailored approaches to activate these cells could enhance their antitumor effects. Therapeutic strategies currently under investigation include reactivating V γ 9V δ 2 T cells with specific antibodies and the adoptive transfer of $\gamma\delta$ T cell subsets. Current trials using V δ 1 and V δ 2 T cells have shown promise in treating solid tumors, hematological malignancies, and bone marrow metastases.^{30–32} Understanding which $\gamma\delta$ T cell subsets naturally home to and persist within tumor beds is critical for developing effective cellular therapies. The database generated by our study provides a valuable resource for advancing these therapeutic strategies, offering avenues for personalized immunotherapy that harness the unique properties of $\gamma\delta$ T cells to improve patient outcomes.

A major analytical challenge of this study arises from the fact that $\gamma\delta$ T cells constitute only a small proportion of lymphocytes, which can complicate their detection and analysis with bulk tumor RNA-seq data. This issue is particularly problematic in cancers with lower immune infiltration, where the presence of $\gamma\delta$ T cells may be even more sparse. Furthermore, the delta chain analysis encounters additional hurdles due to its lesser representation and fewer annotations or references available compared to the gamma chain. Consequently, most of our

analysis focused on the gamma TCR. Even in tumors with higher levels of $\gamma\delta$ T cells, it remains challenging to differentiate the prognostic signature from overall inflammation or immune infiltration, similar to other immune-related signatures. To address this issue, we performed a sensitivity analysis by adjusting for the overall immune infiltration score in the Cox regression model (Figure S6A). Encouragingly, most of the significant genes retained their significance, indicating their prognostic value is at least partially independent of overall immune infiltration.

Another intriguing question warranting further investigation is whether $\gamma\delta$ T cells are more represented in the tumor microenvironment compared to normal tissues. This analysis is challenging with existing data because many normal tissues in cancer genomic datasets are adjacent normal tissues, representing a unique compartment. It is known that $\gamma\delta$ T cells tend to patrol tissues for signs of infection or cellular stress; therefore, they might be enriched in regions closer to tumor. Additionally, normal tissue RNA-seq often receives lower sequencing depth, resulting in a lower detection rate of $\gamma\delta$ T cell-related signatures. Despite these limitations, for the sake of completeness, we analyzed all cancer types with at least five samples with matched normal tissue RNA-seq and found that the diversity of gamma TCRs is significantly higher in breast cancer and lung squamous cell carcinoma (Figure S6B).

This atlas will continue to expand by incorporating more TCR repertoires from published studies, including data from both bulk RNA-seq and single-cell analyses, along with additional functional annotations. Because $\gamma\delta$ T cells are tissue specific and functionally plastic, single-cell RNA-seq would be particularly useful for further investigating their functional roles, cytotoxicity, and effector subsets. Emerging data from tissue-derived $\gamma\delta$ T cells,^{11,33,34} obtained from both animal models and human studies, highlight the importance of such detailed analyses to better understand the diverse roles these cells play in the immune response and their interactions within the tumor microenvironment. This expansion will enable users to efficiently integrate their TCR data with the most up-to-date tumor-specific repertoire available, facilitating a more comprehensive view of the $\gamma\delta$ -T cell-related immune landscape in cancer and potentially uncovering therapeutic avenues.

Limitations of the study

This study has several limitations. First, $\gamma\delta$ T cells constitute a small proportion of lymphocytes, making their detection through bulk RNA-seq challenging, particularly in cancers with low immune infiltration. This may limit the resolution and accuracy of $\gamma\delta$ T cell characterization, especially in tumors with sparse immune cells. Second, our downstream analyses focused more on the gamma chain due to the lower representation of delta chains in the data. This discrepancy may underrepresent the full spectrum of $\gamma\delta$ T cell diversity and function. Additionally, while bulk RNA-seq allows a comprehensive view of $\gamma\delta$ TCRs, it lacks the resolution of single-cell RNA-seq, which could provide more detailed insights into the functional heterogeneity of these cells. Lastly, our study focuses on association and does not establish causality between $\gamma\delta$ T cell features and clinical outcomes, warranting further investigation in prospective studies.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Xuefeng Wang (xuefeng.wang@moffitt.org).

Materials availability

This study did not generate new unique reagents.

Data and code availability

RNA-seq BAM files and FPKM gene expression values were downloaded from Genomic Data Commons (GDC). Clinical information, FPKM values of all $\gamma\delta$ -related genes, and diversity measurements are available at <http://gdt.moffitt.org>. Raw RNA-seq data of eight immunotherapy studies were obtained from Gene Expression Omnibus (GEO) repository with following accession numbers: GSE78220,³⁵ GSE91061,³⁶ GSE121810,³⁷ GSE165252,³⁸ and GSE183924,³⁹ and from European Nucleotide Archive (ENA) with following numbers: PRJEB23709,⁴⁰ PRJEB25780,⁴¹ and PRJNA923698.⁴² Clinical information, processed gene expression in TPM values, and diversity measurements of $\gamma\delta$ TCRs are available at <http://gdt.moffitt.org>. Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

Conceptualization, X.W., X.Y., and L.S.; data curation, X.Y., L.S., L.C., and Y.S.; formal analysis, X.Y., L.S., L.C., and R.T.; software, B.C.; visualization, X.Y. and B.C.; intellectual contributions to study design and review of results, J.R.C.-G., P.C.R., and D.A.-D.; writing – original draft, X.Y., X.W., and L.S.; funding acquisition, X.Y., L.S., and X.W. All the authors reviewed and edited the manuscript.

DECLARATION OF INTERESTS

J.R.C.-G. is a consultant for Anixa Biosciences and Alloy Therapeutics; holds stock options in Compass Therapeutics, Anixa Biosciences, and Alloy Therapeutics; has a patent licensed by Anixa Biosciences; owns intellectual property filed with Compass Therapeutics; and is a co-founder of Cellep Therapeutics, a CAR T cell company.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
STAR	Dobin et al. ⁴³	https://github.com/alexdobin/STAR
TRUST4 v1.0.4	Song et al. ¹⁷	https://github.com/liulab-dfci/TRUST4
FastQC	Andrews et al. ⁴⁴	https://www.bioinformatics.babraham.ac.uk/projects/fastqc/
GSVA	Hänzelmann et al. ⁴⁵	https://github.com/rcastelo/GSVA
<i>ppcor</i>	Kim ⁴⁶	https://cran.r-project.org/web/packages/ppcor/index.html
Cutadapt	Martin ⁴⁷	https://cutadapt.readthedocs.io/en/stable/
RSeQC	Wang et al. ⁴⁸	https://rseqc.sourceforge.net/
Adobe Illustrator 2023	Adobe	N/A
R v4.2.1	R Core Team	https://www.r-project.org/
Deposited data		
TCGA RNA-seq bam files	Genomic Data Commons	https://portal.gdc.cancer.gov/
TCGA patient clinical data	PanCancerAtlas ⁴⁹	N/A
RNA-seq of ICI treated melanoma patients	Hugo et al. ³⁵	GEO: GSE78220
RNA-seq of ICI treated melanoma patients	Riaz et al. ³⁶	GEO: GSE91061
RNA-seq of ICI treated glioblastoma patients	Cloughesy et al. ³⁷	GEO: GSE121810
RNA-seq of ICI treated esophageal adenocarcinoma patients	van den Ende et al. ³⁸	GEO: GSE165252
RNA-seq of ICI treated esophageal and gastroesophageal junction adenocarcinoma patients	Mamdani et al. ³⁹	GEO: GSE183924
RNA-seq of ICI treated melanoma patients	Gide et al. ⁴⁰	PRJEB23709
RNA-seq of ICI treated gastric cancer patients	Kim et al. ⁴¹	PRJEB25780
RNA-seq of ICI treated melanoma patients	Campbell et al. ⁴²	PRJNA923698
Other		
http://gdt.moffitt.org	This manuscript	N/A

EXPERIMENTAL MODES AND SUBJECT DETAILS

Data download and processing

Pre-processed RNA-seq BAM files of 10,970 TCGA samples generated by STAR⁴³ were downloaded from Genomic Data Commons (GDC) through controlled data access application from dbGAP study accession phs00178. Gene counts and Fragments Per Kilobase of transcript per Million (FPKM) expression values (Gencode V36) of TCGA tumors generated by STAR were also obtained from GDC. According to GDC, the raw sequencing reads were first accessed for quality using fastqc and then aligned to GRCh38 with Gencove V36 using STAR two-pass mode, which includes a splice junction detection step. Post-alignment quality assessment was conducted using Picard. Only samples pass the quality criteria of GDC were released. FPKM values of gamma-delta TCR genes we extracted from GDC and used for this study are hosted at <http://gdt.moffitt.org>. Clinical information including overall survival, age, gender, stage, tumor purity of TCGA patients was downloaded from PanCancerAtlas⁴⁹ (TCGA-CDR-SupplementalTableS1.xlsx). A consensus measurement of purity estimations (CPE)⁵⁰ derived from purity level ABSOLUTE, ESTIMATE, LUMP, and IHC was used in downstream analysis. For samples without CPE value available, purity estimated by ABSOLUTE downloaded from PanCancerAtlas (TCGA_mastercalls.abs_tables_JSedit.fixed.txt) were used to represent tumor purity. Microsatellite instability (MSI) scores of COAD and HPV status of HNSC were obtained from cBioPortal. The MSI scores were represented by MANTIS,⁵¹ which was the average values of MSI after calculating the allele distribution differences of each microsatellite site between tumor and samples. COAD tumors were further categorized into MSI-high vs. MSI-low groups, with the cutoff determined by identifying

the local minimum in the bimodal distribution of MSI values density plot (Figure S4E). For HNSC tumors without HPV status available on cBioPortal, we imputed their status based on K-means clustering analysis using the top 100 differentially expressed genes between HPV+ and HPV- tumors.

METHOD DETAILS

TCR reconstruction

The RNA-seq BAM files downloaded GDC from were processed by TRUST4 v1.0.4 to reconstruct immune receptor repertoires in T cells along with their abundances using command line `run-trust4 -f human_IMGT+C.fa -ref human_IMGT+C.fa -b sample.bam -o sample_trust4`. TRUST4 reported complete immune receptor CDR3 sequences, where the sequence started from the IMGT-specified (<https://www.imgt.org/>) V gene location and ended with the IMGT-specified J gene location. The abundance for each CDR3 in this study was estimated using the 3 million candidate TCR reads overlapping CDR3 regions when executing TRUST4. A total of 34,129 CDR3 sequences of gamma-delta ($\gamma\delta$) T cells from 6,751 tumors of 6,392 patients were subjected for analysis. We applied following filtering criteria to remove false positive calls and non-informative TCRs.

- (1) CDR3 sequences from patients lacking associated clinical information were excluded.
- (2) For patients with multiple tumor samples, only the CDR3 sequences assembled from the primary tumor sample with the highest number TCRs identified were retained for the analysis.
- (3) CDR3 sequences carrying alpha-beta V genes were filtered out.
- (4) Out-of-frame CDR3 sequences (459 γ , 8,920 δ) were removed.

To create unique identifiers of $\gamma\delta$ -TCR clones, we used a combination of CDR3 amino acid sequences, V genes, and J genes. Finally, 22,205 distinct $\gamma\delta$ -TCR clones (4,554 γ clones, 17,651 δ clones) from 6,276 tumors across 33 cancer types were retained for repertoire analysis.

Gamma-delta TCR repertoire analysis

Following metrics were calculated to explore repertoire of gamma and delta TCR clones. **Normalized chain count:** For each tumor, gamma (or delta) chain count was calculated as $\log(\frac{r}{R} \times 10000)$, where r is the total count of reads mapped to any gamma (or delta) clones, and R is the total number of RNA-seq reads mapped to transcriptome as in STAR count data downloaded from GDC. **Clone size:** Clone size was defined as total number of reads mapped to each clone. Clone size distribution was analyzed across 33 different cancer types by categorizing clones into sizes of 1, 2, 3–10, 11–100, or >100 reads visualized by `vinplot` function in `ggplot2`. **V-J gene usage:** The usage of V-J gene pairs was explored by calculating the number of distinct clones (counts) and percentage of CDR3 reads (frequency) harboring a specific combination of V-J pair. V-J pairs with a frequency higher than 0.3% were visualized in stacked bar plots for each cancer type using `ggplot2`. **CDR3 length:** CDR3 amino acid length distribution analysis was conducted across 33 cancer types in gamma and delta clones respectively and visualized in ridge plot in `ggplot2`. Cancer types were ranked by the mode (max values) of CDR3 length density distribution. **Clonality and entropy:** For each tumor with at least 5 distinct clones, the clonality of gamma and delta clones was measured by Shannon clonality ($1 - \frac{\sum_{i=1}^N p_i \log(p_i)}{\log N}$) and Simpson clonality ($\frac{\sum_{i=1}^N p_i^2}{\sum_{i=1}^N p_i}$), where p_i is the frequency of clone i , and N is the total number of distinct clones. We also calculated normalized Shannon entropy as $\frac{\sum_{i=1}^N p_i \log(p_i)}{\log N}$ to measure diversity. Clonality close to 0 characterizes a polyclonal repertoire with equivalent representations of each clone (i.e., infinite diversity); clonality close to 1 represents a highly oligoclonal repertoire (i.e., no diversity). We compared the clonality and entropy of gamma-delta clones between HPV-positive and HPV-negative HNSC tumors and between MSI-high and MSI-low COAD tumors using the Wilcoxon signed-rank test (Figures 4, 5, and S4). To further investigate gamma-delta T cell infiltration differences in tumor vs. normal samples, we compared normalized Shannon entropy between matched tumor-normal pairs in TCGA (Figure S6B).

Gamma-delta TCR gene expression

The overall enrichment of gamma (or delta) gene expression was calculated from TPM values downloaded from GDC using single-sample GSEA implemented in R package `GSVA`⁴⁵ on all TCR gamma (or delta) genes with default setting. In order to correlate normalized gamma (or delta) TCR clone abundance with gamma (or delta) gene expression enrichment scores in each cancer type, partial Spearman rank correlation controlling for tumor purity of each tumor was performed using `pcore.test` function in `ppcor` R package⁴⁶ with method = "spearman". The prognostic potentials of individual gamma delta TCR genes (V, D, J, C) expression were evaluated by Cox proportional hazard regression analysis on TPM values adjusted for covariates including gender, age, stage, and tumor purity across different cancer types, using `coxph` function of R package `survival`. p values were further adjusted by the Benjamini-Hochberg method to correct for multiple comparisons. The Hazard Ratio and confidence interval were visualized for gamma-delta TCR genes with adjusted p -value <0.05. We also examined the correlation of the expression of individual gamma delta TCR genes and TCR enrichment scores with different tumor subtypes such as: HPV-positive vs. HPV-negative HNSC tumors, and MSI-high vs. MSI-low

COAD tumors. Specifically, TCR gene TPM values and enrichment scores were first adjusted for tumor purity by taking residuals from linear regression; then Wilcoxon signed-rank test was performed to compare the purity adjusted values between HPV-positive vs. HPV-negative HNSC tumors, as well as between MSI-high vs. MSI-low COAD tumors; results were visualized by boxplot function in *ggplot2*. In addition, to assess differences in the prognostics potentials of expression levels of individual TCR genes among above different tumor subtypes, we conducted Cox proportional hazard regression adjusting for gender, age, stage, and tumor purity using *coxph* function of R package *survival*. *p* values were further adjusted by the Benjamini–Hochberg method. The Hazard Ratio (and confidence interval) and adjusted *p*-values were visualized for genes showing different prognostics values between different tumor subtypes for HNSC and COAD. Similarly, cox regression adjusted for gender, age, stage, and tumor purity was performed and visualized to evaluate the different prognostics potentials of TCR enrichment scores between HPV-positive vs. HPV-negative HNSC tumors, as well as between MSI-high vs. MSI-low COAD tumors. To differentiate the prognostic signature from overall inflammation or immune infiltration, we also performed a sensitivity analysis by adjusting for the overall immune infiltration score (IIS) in the Cox regression model (Figure S6A).

Data analysis of immunotherapy studies

RNA-seq reads of eight immunotherapy studies were downloaded from GSE78220 (Hugo et al.³⁵), GSE91061 (Riaz et al.³⁵), GSE121810 (Cloughesy et al.³⁷), GSE165252 (van den Ende et al.³⁸), GSE183924 (Mamdani et al.³⁹), PRJEB23709 (Gide et al.⁴⁰), PRJEB25780 (Kim et al.⁴¹), and PRJNA923698 (Campbell et al.⁴²). The raw sequencing reads were first assessed for quality using FastQC (<http://www.bioinformatics.babraham.ac.uk/projects/fastqc/>). Quality trimming were performed using cutadapt⁴⁷ to remove reads with adaptor contaminants and low-quality bases. Reads pairs with either end too short (<25bps) were discarded from further analysis. Strandness of reads was inferred by RSeQC *infer_experiment.py*⁴⁸ with default setting. Next, trimmed and filtered reads were aligned to the human transcriptome GRCh38 using STAR⁴³ with parameters `–outSAMtype BAM SortedByCoordinate –read-FilesCommand zcat –twopassMode Basic –chimOutType Junctions SeparateSAMold –outFilterMismatchNoverLmax 0.04 –chim-SegmentMin 10 –quantMode TranscriptomeSAM`. Gene-level TPM (transcripts per million) values were obtained by RSEM⁵² using Gencode V30 with parameters `–star –star-gzipped-read-file –estimate-rspd –strandedness S`, where S is strandness inferred by RSeQC. Overall enrichment of gamma (or delta) gene expression was then calculated from TPM values using ssGSEA as described above. Wilcoxon signed-rank test was performed to compare gamma (or delta) enrichment between responders vs. non-responders, and pre-treatment vs. after-treatment. For Riaz study, we split the patients into ipilimumab treatment naive (Ipi Naive) and progressive (Ipi Prog) following the original paper. For studies with survival data available, Kaplan-Meier survival analysis was performed to compare overall survival between groups using optimal gamma (or delta) enrichment as cutoff. The optimal cutoff was determined using *surv_cutpoint* function in R package *surminer*. The prognostic potentials of individual gamma delta TCR genes (V, D, J, C) expression were evaluated by Cox proportional hazard regression analysis on TPM values adjusted for gender and age using *coxph* function of R package *survival*.

Immune receptor repertoires of T cells were constructed from the raw sequencing reads using TRUST4 v1.0.4 with command line as following. Paired-end data: `run-trust4 -f human_IMGT+C.fa –ref human_IMGT+C.fa –1 sample_1.fastq –2 sample_2.fasta -o sample_trust4`; Single-end data: `run-trust4 -f human_IMGT+C.fa –ref human_IMGT+C.fa –1 sample.fastq -o sample_trust4`. CDR3 sequences and $\gamma\delta$ -TCR clones were identified and filtered as described above in TCR reconstruction section for TCGA. A total of 3,574 $\gamma\delta$ -TCR clones (2,887 γ clones, 687 δ clones) from 469 tumors across 8 immunotherapy studies were retained for repertoire analysis. Three metrics were calculated to measure diversity of gamma-delta TCR repertoire: True diversity (effective number of species), $(\sum_{i=1}^N p_i)^{1/(1-q)}$; Normalized Shannon entropy, $\frac{\sum_{i=1}^N p_i \log(p_i)}{\log N}$; Inverse Simpson index, $\frac{1}{\sum_{i=1}^N p_i^2}$, where p_i is the frequency of clone i , N is the total number of distinct clones, and $q = 5$. Wilcoxon signed-rank test was performed to compare gamma (or delta) diversity between responders vs. non-responders, and pre-treatment vs. after-treatment. Kaplan-Meier survival analysis was performed to compare overall survival between groups with high vs. low true diversity, with optimal cutoff determined using *surv_cutpoint* function in R package *surminer*.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed in R version 4.2.1. All plots were generated using the *ggplot2* package in R with the exception that the Kaplan-Meier curves were generated using *surminer* package. Statistical comparisons of two groups utilized two-sided Wilcoxon rank-sum test and visualized by violin plot or boxplot with median represented by central line, the 25% and 75% interquartile (IQR) represented by lower and upper hinges, and the ± 1.5 IQR represented by Tukey whiskers; $p < 0.05$ was considered to be statistically significant. Differences in survivals were calculated by Cox proportional hazards regression and results were visualized as Kaplan-Meier curves or forest plots. Spearman correlation was calculated using *cor()* function in R. Statistical details can also be found in the method details and figure legends.